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EXPERIMENT 2 – STUDY OF NETWORK CABLES

AIM

To understand the construction, characteristics, and applications of commonly used computer network cables and to gain practical knowledge of preparing straight-through and cross-over Ethernet cables using RJ45 connectors and a cable crimping tool.

REQUIREMENTS

- CAT5e Ethernet cable with four twisted wire pairs
- RJ45 modular connectors
- RJ45 cable crimping tool
- LAN cable tester
- Wire cutter or cable cutting tool
- Wire stripper
- Network devices such as a computer, switch, or router for testing
- A suitable workspace for arranging and terminating the cable

TYPES OF NETWORK CABLES

1. UTP – Unshielded Twisted Pair

Unshielded Twisted Pair cable consists of multiple pairs of insulated copper wires twisted around each other. The twisting arrangement helps reduce electromagnetic interference and crosstalk between neighboring wire pairs. Since the cable does not contain an additional metallic shield, it is lightweight, flexible, and relatively inexpensive.

UTP cables are widely used in Local Area Networks (LANs), offices, computer laboratories, and residential networks. Different categories such as CAT5e, CAT6, and CAT6a provide different levels of network performance and transmission capability.

2. STP – Shielded Twisted Pair

Shielded Twisted Pair cable contains twisted copper wire pairs along with a conductive shielding layer. The shielding provides additional protection against electromagnetic interference from electrical equipment and other sources of noise.

STP cables are useful in environments where there may be considerable electrical interference. They are commonly considered for industrial areas, communication rooms, and locations containing heavy electrical machinery. Compared with UTP, STP cables are generally thicker, less flexible, and require appropriate grounding for effective shielding.

3. Coaxial Cable

Coaxial cable contains a central conducting wire surrounded by an insulating layer, a metallic shield, and an external protective covering. Its layered construction provides good resistance to external interference and helps maintain signal quality.

Coaxial cables have been used in older computer networks and are still found in applications such as cable television, broadband connections, CCTV systems, and other communication systems. Although modern LANs commonly use twisted-pair or fibre-optic connections, coaxial cable remains useful in several communication applications.

4. Fibre Optic Cable

Fibre optic cable transfers information in the form of pulses of light through very thin strands of glass or plastic. It provides high bandwidth and can transmit signals over much longer distances than conventional copper Ethernet cables.

Since fibre optic communication uses light instead of electrical signals, it is not affected by electromagnetic interference in the same way as copper cables. Fibre optic cables are therefore widely used for backbone networks, data centres, telecommunications, and high-speed network connections.

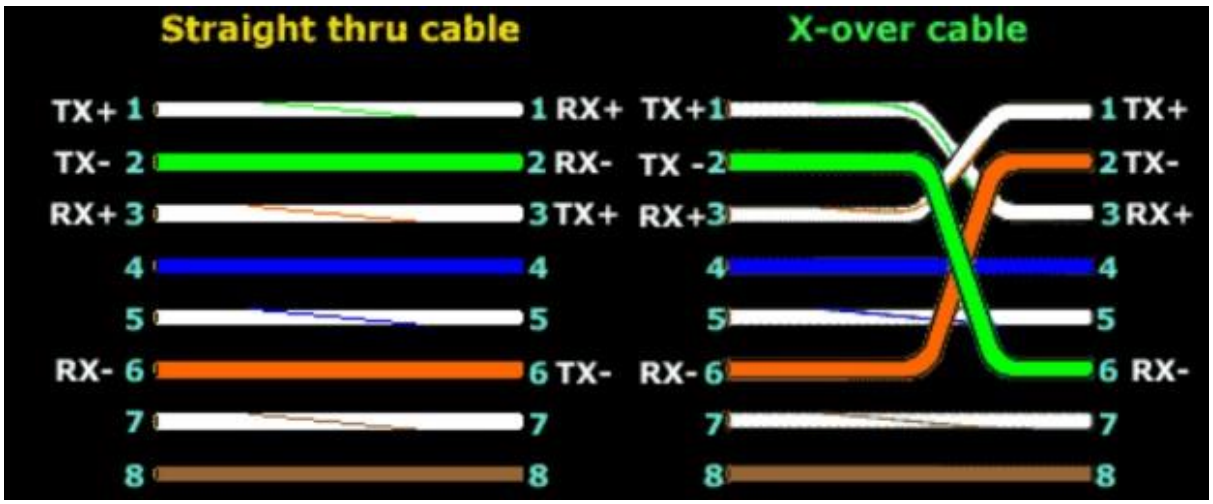
PROCEDURE

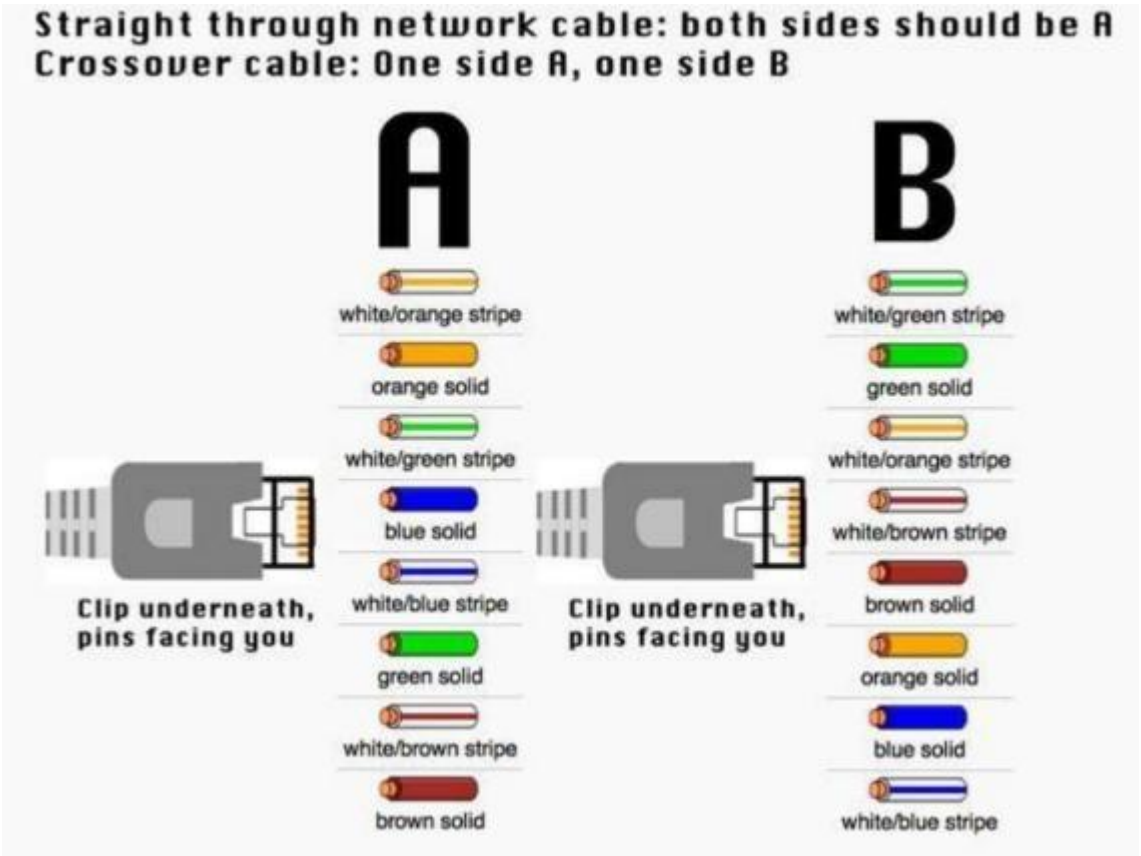
1. Determine the required length of Ethernet cable according to the intended network connection and cut the cable using a suitable cutter.

2. Carefully remove a small portion of the outer jacket from both ends of the cable without damaging the individual insulated wires inside.
3. Identify the four twisted pairs present inside the Ethernet cable and gently separate them.
4. Straighten the individual conductors so that they can be placed in the required sequence.
5. Select the appropriate Ethernet wiring standard for terminating the cable. The commonly used standards are T568A and T568B.
6. For preparing a straight-through cable, arrange the conductors using the same wiring standard at both ends of the cable.
7. For preparing a cross-over cable, terminate one end using T568A and the other end using T568B.
8. After arranging the conductors, hold them firmly together and trim their ends so that they are approximately equal in length.
9. Insert the arranged conductors into the RJ45 connector. Make sure that each wire enters its corresponding channel and reaches the front portion of the connector.
10. Check the connector carefully to ensure that the wires remain in the correct sequence and that the outer cable jacket enters the connector sufficiently to provide mechanical support.
11. Place the RJ45 connector inside the crimping tool and apply firm pressure so that the connector contacts are pressed onto the individual conductors.
12. Repeat the termination process for the opposite end of the cable using the selected wiring arrangement.
13. Visually inspect both connectors for loose wires, incorrect ordering, or improper crimping.
14. Connect the completed cable to an Ethernet cable tester and verify the continuity of each conductor.

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15. If the tester indicates the correct sequence and continuity, connect the cable between suitable networking devices and verify that the network link is established.





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OBSERVATION

Cable Type	Wiring Arrangement	Typical Connection	Main Characteristic
Straight-through cable	Same standard at both ends	Computer to Switch/Router	Used for connecting different types of network devices
Cross-over cable	T568A at one end and T568B at the other	Computer to Computer	Used for direct connections between similar network devices in situations where auto-sensing is unavailable

RESULT

Thus, the major types of network cables were studied along with their construction, characteristics, and applications. A straight-through Ethernet cable and a cross-over Ethernet

cable were prepared by arranging the conductors according to the required wiring standards and terminating them with RJ45 connectors. The completed cables were checked using a cable tester to verify proper continuity and wiring.